

BINOCULAR ASTRONOMY

64–67	Celestial cycles
76–77	Naked-eye astronomy
82–85	Telescope astronomy
348–53	Mapping the sky
426–501	Monthly sky guide

FOR MOST NEWCOMERS to astronomy, the most useful piece of equipment is a pair of binoculars. As well as being easy and comfortable to use, binoculars (unlike telescopes) allow stargazers to see images the right way up. A range of fascinating astronomical objects can be observed through them.

BINOCULAR CHARACTERISTICS

Binoculars are like a combination of two low-powered telescopes. The two main designs, called Porro-prism and roof-prism, differ in their optics, but either can be useful for astronomy. More important when choosing binoculars are the two main numbers describing their optical qualities; for example, 7×50 or 12×70. The first figure is the magnification. For a beginner, a magnification of 7× or 10× is usually adequate—with a higher magnification, it can be difficult to locate objects in the sky. The second figure is the aperture, or diameter of the objective lenses, measured

in millimeters. This number expresses the binoculars' light-gathering power, which is important in seeing faint objects.

For night-sky viewing, an aperture of at least 50 mm (2 in) is preferable.



STANDARD BINOCULARS

These typically have 50-mm (2-in) objective lenses and a magnification of 7× or 10×. This pair has a Porro-prism design.



COMPACT BINOCULARS

These are lightweight but their objective lenses are rather small for astronomy. This pair has a roof-prism design.

EXPLORING SPACE

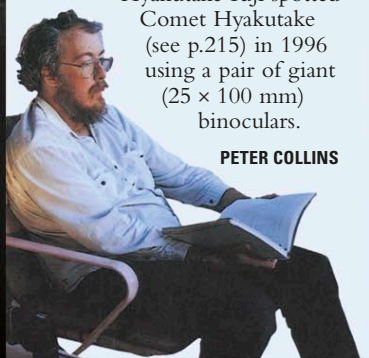
BINOCULAR FINDS

Astronomers make some important discoveries using binoculars.

Arizona astronomer Peter Collins uses binoculars to search for the stellar outbursts known as novae (see p.282). To make the method effective, he memorizes thousands of star positions. Comets are also frequently first seen by binocular enthusiasts.

Japanese astronomer Hyakutake Yuji spotted Comet Hyakutake (see p.215) in 1996 using a pair of giant (25 × 100 mm) binoculars.

PETER COLLINS



IDYLIC SKYGAZING

The modest magnifying power of binoculars is more than enough to reveal many of the sky's most interesting objects. Wilderness camping is a good way to get away from light pollution.

